

Original Research

Effect of Oil Palm Broom Fiber on the Mechanical Properties of Rice Husk Ash–Blended Concrete

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Abstract

The high carbon footprint of cement production and the cost and environmental impact of steel reinforcement make conventional concrete unsustainable for long-term use. Reducing cement consumption and identifying sustainable, cost-effective alternatives to steel are critical for lowering construction's ecological and economic burden. Bio-based supplementary cementitious materials (SCMs), such as rice husk ash (RHA), and natural plant fibers show promise, but prior studies have mostly tested them separately or with varying combinations, leaving the specific effect of fiber dosage at fixed SCM levels remain unclear. Therefore, this study addresses this gap by systematically investigating the effect of oil palm broom fiber (OPBF) content (0–5% by weight, in 0.5% increments) on the fresh and hardened properties of concrete containing a fixed 10% RHA replacement of cement. The physical, fresh and mechanical properties of OPBF were evaluated, and statistical analyses (ANOVA, Tukey's HSD) were conducted to identify significant performance differences and determine the optimal fiber content. Experimental results indicate that increasing OPBF content reduced workability



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by up to 62.5% observed at 5% OPBF, while compressive and flexural strengths peaked at 2.5% (27.8 MPa) and 3.0% OPBF (5.8 MPa), respectively. The results go beyond previous efforts by providing the first systematic dataset on OPBF reinforced RHA-blended concrete performance, establishing optimum fiber dosages, and demonstrating the potential of agricultural residues to produce sustainable, performance-optimized concrete for low-load-bearing applications.

Keywords

Oil palm broom fiber (OPBF); rice husk ash (RHA); ANOVA; Tukey's HSD; mechanical properties

1. Introduction

Concrete, the most familiar manmade construction material, is vastly consumed due to the advantages it possesses, which include serviceability, long-term durability, availability, and cost-effectiveness of its constituent materials [1-3]. It has been utilized in various types of construction projects, including residential buildings, skyscrapers, bridges, dams, and pavements [4]. It is estimated that global consumption of concrete might reach 30 billion tons annually, which can be derived from the values of annual cement production reaching 4.0 billion tons by 2030 [5]. However, by the early 1800s, it was understood that concrete is generally brittle in character, has low energy absorption capacity after yielding, and possesses poor tensile resistance [6, 7]. To eliminate these limitations and make concrete tougher and more ductile, the use of steel as reinforcement gained significant attention. Steel has been commonly used as longitudinal reinforcement and, in recent times, as randomly dispersed fibers [8-11]. However, from both an economic and environmental perspective, the combination of concrete and steel is both detrimental and expensive, as concrete and rebar together account for 65% of construction-related greenhouse gas emissions, with rebar alone contributing 60% of this total [12], and their combined use becomes expensive when rebar is misused or wasted beyond actual requirements [13].

Furthermore, Concrete production is reported to be a major contributor to GHG emissions [14, 15]. The most important constituent material of concrete, Ordinary Portland Cement (OPC), has been reported by several researchers to be responsible for 5–8% of global CO₂ emissions [16-22]. Similarly, steel production is another major contributor to global CO₂ emissions. In fact, 7–8% of global CO₂ emissions are attributed to steel production [23], and while rebar production contributed to the steel industry's CO₂ emissions, calculating its exact share requires more detailed data. In addition to these detrimental environmental impacts, the continuous increase in infrastructural costs due to the reliance on outdated steel production processes poses a persistent challenge for the governments of developing countries in providing affordable and decent housing [24-27]. As a result, the search for alternative construction materials that are more cost-effective to produce, readily available, satisfies load bearing requirements and, most importantly, environmentally sustainable has gained significant attention in recent times.

The combined incorporation of natural fibers and supplementary cementitious materials (SCMs) derived from agricultural waste presents a promising and sustainable approach to reinforced concrete production. This strategy not only offers a means of reducing the environmental burden

associated with conventional cement and steel but also holds potential for improving the mechanical performance of concrete, particularly in resource-constrained regions. Over the past decade, there has been a growing interest in utilizing bio-based SCMs (such as rice husk ash (RHA), palm oil fuel ash (POFA)) and natural fibers (such as coconut coir, jute, and hemp) to partially or fully replace traditional materials [28-33]. However, a closer look at the existing literature reveals that most studies have either focused on the combination of bio-based SCMs with synthetic fibers such as polypropylene and steel [34-37], or on the incorporation of natural fibers with industrial byproduct-based SCMs (such as fly ash and slag) [38-40]. While these efforts demonstrate the potential of hybrid sustainable mixes, there remains a noticeable gap in research that simultaneously evaluates the combined effect of bio-based SCMs and natural plant fibers on concrete performance. In particular, studies that integrate materials such as RHA with underutilized natural fibers like oil palm broom fiber (OPBF) are rare. Furthermore, many investigations have assessed these materials in isolation, examining either SCMs or fibers alone, rather than exploring their combined behavior [41-43].

Several previous studies have also investigated RHA with plant-based fibers, but their scopes differ significantly. Raheem *et al.* (2021) examined thermal insulation properties of concrete made with palm kernel shell and RHA [44], while Aminu and Sharma (2022) studied mechanical properties of recycled aggregate concrete using hybrid steel–glass fibers with RHA [45]. Wahyuni *et al.* (2014) investigated bamboo fiber-reinforced concrete incorporating RHA and seashell ash but did not isolate the effects of fiber dosage in an RHA-based binder [46]. In contrast, this study addresses the underexplored role of fiber content in concrete with a fixed RHA replacement level, specifically using OPBF. While RHA and plant-based fibers have been studied, no study, to the best of our knowledge, has examined how varying OPBF content influences both fresh and hardened properties when RHA is held constant. Addressing this gap is essential for advancing the development of performance-optimized concrete using agricultural residues.

To inform this investigation, a focused literature review on RHA-based and fiber reinforced concrete was conducted. For instance, Tariq *et al.* reported that concrete mixes with up to 18% RHA replacement exhibited minimal reductions in compressive, splitting, and flexural strength compared to control specimens. Moreover, the incorporation of RHA was shown to enhance the cost-efficiency and eco-friendliness of the concrete mix, reinforcing its potential as a viable bio-based SCM [41]. Additionally, incorporating RHA as a SCM made the mix more cost-effective and environmentally friendly. Similarly, Muleya *et al.* found that partially replacing cement with up to 30% RHA provided economic benefits while producing high-quality, practical concrete. Their study highlighted the potential of RHA in improving the standard of living for rural communities in rice-growing regions [42]. Zaid *et al.* observed that a 10% replacement of cement with RHA had no significant impact on concrete performance. They further noted that adding steel fibers and admixtures, such as superplasticizers, could enhance the economic viability of concrete for the construction industry [47]. Jhatial *et al.* investigated the effects of steel fibers on the mechanical properties of concrete and found that their inclusion improved compressive and flexural strength while enhancing ductility and crack resistance. However, increasing the steel fiber content led to a reduction in workability [48]. Yaswanth and Premkumar examined the behavior of concrete reinforced with bristle coir fibers and reported that their inclusion enhanced compressive, split tensile, and flexural strength. The maximum strength was achieved at 1.5% fiber content by total volume of concrete [49]. Similarly, Sohu *et al.* found that while adding plastic fibers reduced

concrete workability, it increased flexural strength by up to 16.5% compared to the control sample [50].

The studies discussed above highlight the importance of optimizing SCM and fiber content in concrete to achieve desired structural outcomes. In this context, North America has an abundant supply of RHA, which presents a promising opportunity for sustainable concrete production. As of 2023, America's annual rice output had reached over 9.9 million metric tonnes [51]. In addition to this, there is an abundance of natural fibers originating from various trees, among which OPBF has recently gained attention as a potential reinforcing fiber for concrete. Because of their remarkable mechanical qualities, the ribs of the oil palm tree's leaflets (also referred to as OPBF) have recently drawn the attention of researchers [52-54] and have used it as reinforcing fibers in plain cement concrete. Unlike other vegetable fibers, OPBF has a low affinity for water, does not decay readily, and has a tensile strength-to-weight ratio that is around five times that of steel [54].

Given the availability of these agricultural by-products and the need for resource-efficient construction materials, this study investigates the effect of OPBF content on the mechanical performance of concrete incorporating 10% RHA as a partial cement replacement. Concrete mixes were prepared with OPBF contents ranging from 0% to 5%, and evaluated for workability, compressive strength, and flexural strength. Additionally, the physical and mechanical properties of OPBF were characterized, and statistical analysis was performed using one-way ANOVA and Tukey's Honest Significant Difference (HSD) test to identify significant performance differences.

2. Materials and Methods

2.1 Materials

Oil Palm Broom Fiber (OPBF), Rice husk ash (RHA), cement, sharp sand, granite, water and superplasticizer were used for concrete production.

The OPBF was sourced from Ipata Market in Ilorin, Nigeria. The diameter of the OPBF, measured using a digital Vernier caliper, ranged from 1.8 mm to 2.0 mm. The fiber length for this research was set at 45 mm as shown in Figure 1(a), maintaining an average aspect ratio of 22.5. Physical and mechanical properties, including moisture content, water absorption, and tensile strength of the OPBF, were evaluated. The findings from these tests are thoroughly discussed in the results and discussion section of this article.

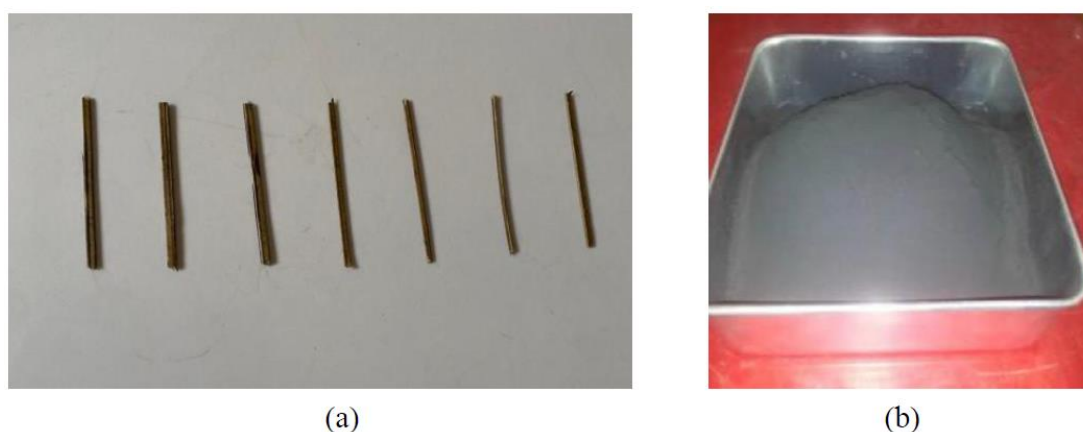


Figure 1 Appearance of (a) OPBF, (b) RHA.

The rice husk was collected from a rice milling company located in the Ojagboro area of Ilorin, Nigeria. A controlled incineration chamber at a Fabrication Workshop in Ilorin, Nigeria, was used to calcine the rice husk. RHA was obtained by subjecting the rice husk to controlled combustion at 700°C for four hours. After combustion, the grey-colored RHA was produced, ground for a few minutes, and passed through a 0.6 mm sieve. The appearance of RHA was shown in Figure 1(b), and its chemical composition was determined using X-ray fluorescence (XRF) with a measurement uncertainty of ± 0.5 wt% (weight percent) for major oxides, ± 0.1 – 0.3 wt% for minor oxides, and ± 0.2 wt% for loss on ignition (LOI), as confirmed by triplicate measurements. The results of the chemical analysis of RHA are summarized in Table 1. The combined percentage of Silicon Oxide, Iron Oxide, and Aluminum Oxide was found to be 80.86%, exceeding the 70% threshold. According to BS EN 450-1 [55], this confirms that RHA is a good pozzolanic material. However, it is important to note that while XRF identifies elemental composition, it does not determine the mineralogical phase of silica such as cristobalite, to confirm the amorphous nature of the silica and validate pozzolanic reactivity more conclusively.

Table 1 Chemical Compositions of RHA.

Chemicals	SiO ₂	Fe ₂ O ₃	Al ₂ O ₃	Na ₂ O	CaO	MgO	K ₂ O	SO ₃	LOI
RHA (%)	63.80	6.32	10.74	1.62	9.78	3.81	3.26	0.06	5.48

An Ordinary Portland Cement (OPC) of grade 42.5R, manufactured by Dangote Cement Company, was sourced from a local cement distribution store in Nigeria. The fine and coarse aggregate (FA and CA) used for this experiment were procured from BUA construction site at the University of Ilorin, Nigeria. The FA is from sharp sand, while the CA is from granite, with sizes ranging between 5 mm and 20 mm. The physical properties of FA and CA are summarized in Table 2.

Table 2 Physical properties of aggregates.

Property	Fine aggregate	Coarse aggregate	Standard followed
Specific gravity	2.66	2.77	
Water absorption capacity, %	1.21	0.91	
Moisture content, %	1.63	4	BS EN 12620 [56]
coefficient of uniformity (Cu)	3	2.43	
coefficient of curvature (Cc)	1.33	1.42	

The particle size distribution of the aggregates was first determined in accordance to BS EN 933 [57]. The sand is classified as poorly graded with a coefficient of uniformity (Cu) of 3.0, a coefficient of curvature (Cc) of 1.33, and an average specific gravity of 2.66. Similarly, the granite exhibits a coefficient of uniformity (Cu) of 2.43 and a coefficient of curvature (Cc) of 1.42, indicating that the granite is also poorly graded. Water absorption of OPBF was assessed by soaking pre-weighed fibers in water at room temperature for 24 hours, followed by blotting and re-weighing to determine absorbed moisture content.

Tap water which conforms to the requirements of the BS EN 1008 [58] was used. AXION TUFFCRETE liquid polymer, in accordance with the manufacturer's specifications, was used as a superplasticizer to enhance the flow of the concrete mix when OPBF was added.

2.2 Mixture Proportion and Making Cube and Prism Specimens

In this experimental program, a series of concrete mixtures were designed by varying the OPBF percentages (0%, 0.5%, 1.0%, 1.5%, 2.0%, 2.5%, 3.0%, 3.5%, 4.0%, 4.5%, and 5.0%). The 0.5% increment was chosen with the consideration that a wide range of variations might ensure a measurable difference between the discrete mixes for a particular parameter, allowing for a meaningful statistical analysis. The amount of cement replaced with RHA was kept constants for all cases. In a previous study, Silva and Naveen examined the effect of RHA and coconut coir fiber on the sustainability of mortar production and suggested that adding more than 15% RHA is not desirable as it reduces the flexural strength [59]. The concrete mix design was carried out in accordance with the Department of Environment (DOE) method, targeting a characteristic strength of 25 MPa at 28 days. Table 3 summarizes the mixture proportion of concrete, and the experimental workflow is shown in Figure 2. The water-to-binder (W/B) ratio was 0.5 for all mixtures, and the superplasticizer was used at 1% by weight of the binder. The materials were batched by weight and mixed thoroughly to achieve homogenous concrete paste and concrete cubes of 100 mm × 100 mm × 100 mm sizes and concrete prism of 100 mm × 100 mm × 500 mm sizes were cast as samples for each concrete mix as shown in Figure 3(a) and (b). The samples were cured inside water at room temperature ($23 \pm 2^\circ\text{C}$) for 7, 14 and 28 days as per **BS EN 12390** [60].

Table 3 Concrete mixture proportions.

MIX	CEMENT (Kg/m ³)	RHA (Kg/m ³)	SAND (Kg/m ³)	GRANITE (Kg/m ³)	WATER (Kg/m ³)	SP (Kg/m ³)	OPBF (Kg/m ³)	REMARKS
MIX 1	378.0	42.0	698.0	1162.0	210.0	4.20	0.00	Control - 10% RHA + 0% OPBF
MIX 2	378.0	42.0	698.0	1162.0	210.0	4.20	2.10	10% RHA + 0.5% OPBF
MIX 3	378.0	42.0	698.0	1162.0	210.0	4.20	4.20	10% RHA + 1.0% OPBF
MIX 4	378.0	42.0	698.0	1162.0	210.0	4.20	6.30	10% RHA + 1.5% OPBF
MIX 5	378.0	42.0	698.0	1162.0	210.0	4.20	8.40	10% RHA + 2.0% OPBF
MIX 6	378.0	42.0	698.0	1162.0	210.0	4.20	10.50	10% RHA + 2.5% OPBF
MIX 7	378.0	42.0	698.0	1162.0	210.0	4.20	12.60	10% RHA + 3.0% OPBF
MIX 8	378.0	42.0	698.0	1162.0	210.0	4.20	14.70	10% RHA + 3.5% OPBF
MIX 9	378.0	42.0	698.0	1162.0	210.0	4.20	16.80	10% RHA + 4.0% OPBF
MIX 10	378.0	42.0	698.0	1162.0	210.0	4.20	18.90	10% RHA + 4.5% OPBF
MIX 11	378.0	42.0	698.0	1162.0	210.0	4.20	21.00	10% RHA + 5.0% OPBF

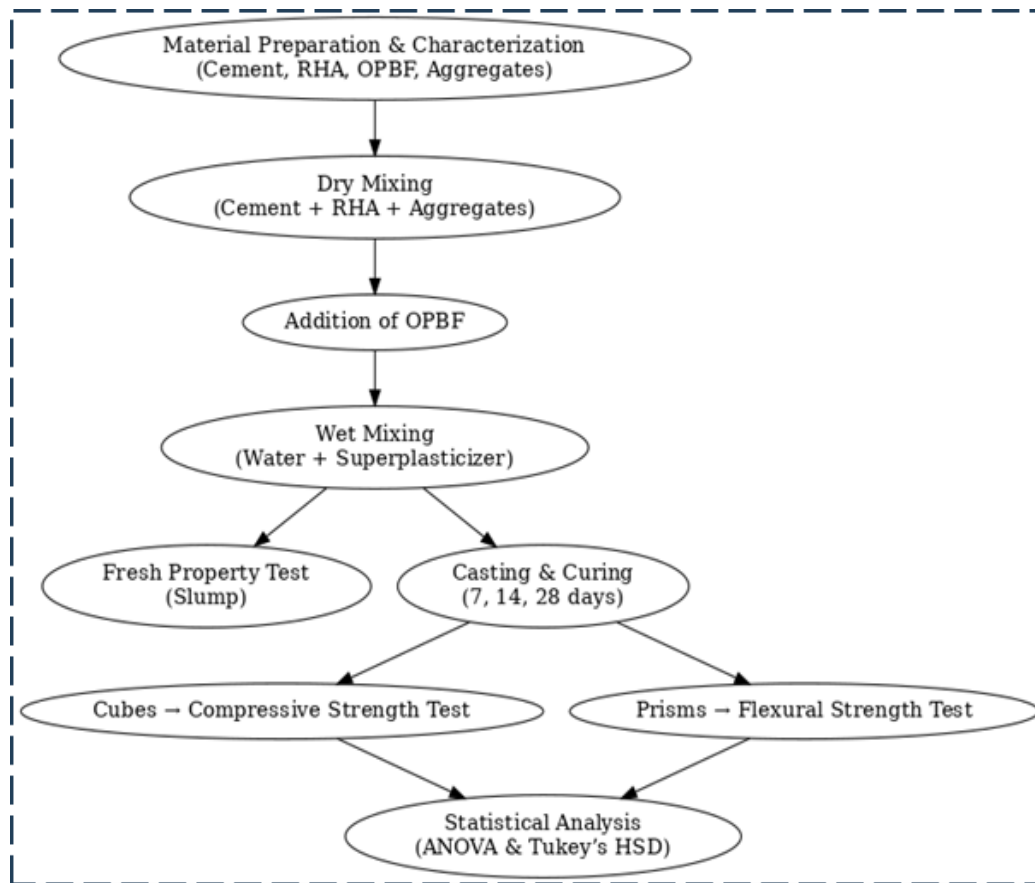


Figure 2 Experimental workflow of the study.

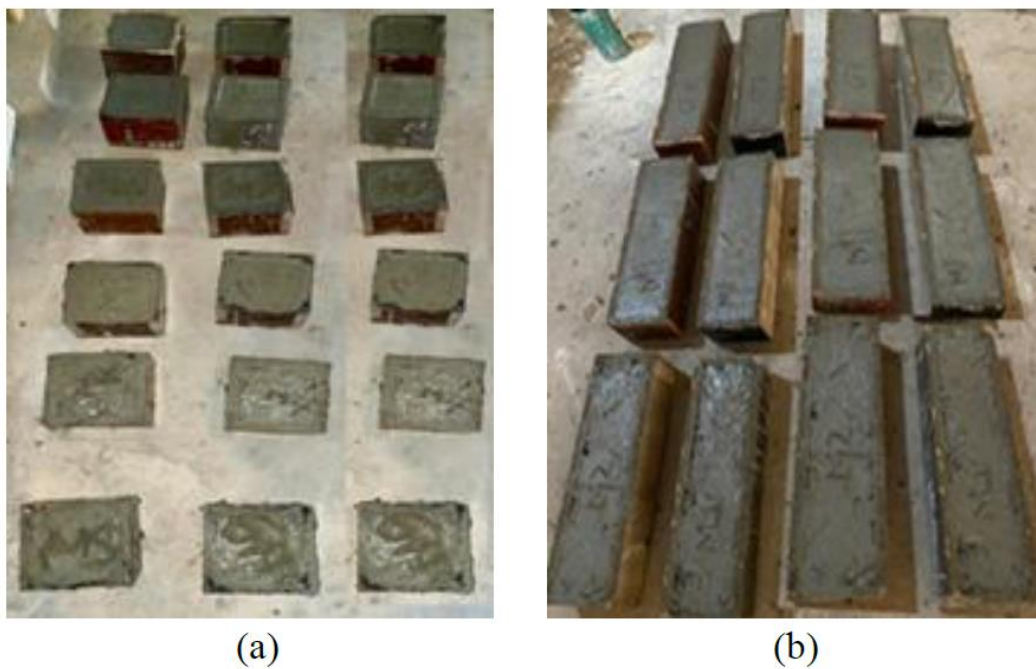


Figure 3 Casting of concrete (a) cubes (b) prisms.

2.3 Test Procedure for Workability, Compressive Strength and Flexural Strength

The workability of the fresh concrete for each mix was determined in accordance with BS EN 12350-2 [61].

At the ages of 7, 14, and 28 days, concrete cube specimens were tested for compressive strength as per BS EN 12390-3 [62]. For a specific day, multiple concrete cubes (per case) were tested, and the average value was considered the final result. Error bars, based on the standard deviation calculated from the concrete cube data points, were used to show the spread of data around the average and to visualize the variability among the test results.

At the ages of 7 and 28 days, flexural strength was determined using a three-point bending test in accordance with BS EN 12390-5:2019 [63]. Prismatic concrete specimens with nominal dimensions of 100 mm × 100 mm × 500 mm were tested using a universal testing machine. The support span was set at 300 mm, and the load was applied at a constant rate of 100 N/s, which falls within the acceptable range specified by the standard. For a specific day, multiple concrete prisms (per case) were tested, and the average value was considered the final result. Error bars, based on the standard deviation calculated from the concrete prism data points, were used to show the spread of data around the average and to visualize the variability among the test results.

3. Results and Discussion

3.1 Physical and Mechanical Properties of OPBF

Table 4 summarizes the physical and mechanical properties test result conducted on OPBF. The average fiber diameter was measured to be approximately 2.0 mm, with values ranging from 1.8 mm to 2.2 mm based on multiple measurements along the length of representative fibers. The moisture content of OPBF used in this study was determined to be 12.44%, which is slightly higher than the value (9.86%) found by Momoh *et al.* [52] for OPBF that was 365 days old. The slightly higher moisture content can be attributed to the possibility that the OPBF used in this study was harvested relatively recently, allowing less time for natural moisture evaporation before testing. However, it is crucial to maintain the moisture content below 10% to prevent dynamic instability of the fibers caused by moisture evaporation, ensuring that the fiber-matrix bond remains unaffected [52].

Table 4 Physical and Mechanical properties of OPBF.

Parameters	Moisture Content (%)	Water Absorption (%)	Mean Diameter, D (mm)	Mean Length, L (mm)	Aspect Ratio, (L/D)	Youngs Modulus (MPa)	Tensile Strength (MPa)	Strain @peak (%)
OPBF	12.44	29.54	2.0	45	22.5	15774.04	105.38	1.36

The water absorption was found to be 29.54%, which is much lower than the value (44.7%) reported by Momoh *et al.* [52]. Moreover, OPBF exhibits considerably lower water absorption than several other natural fibers, including banana fiber, cork fiber, and eucalyptus fiber [64]. The observed absorption capacity may relate to the fiber's surface morphology and internal structure. While previous studies have reported differing values [65], these variations could be due to differences in fiber treatment, anatomical origin, or regional characteristics.

The tensile strength of OPBF was determined to be 105.38 MPa. Compared to previously reported values of tensile strength (300 MPa – 900 MPa) [52], the OPBF used in this study exhibited considerably lower tensile strength. The possible reason can be attributed to the high moisture content of the OPBF used. It is likely that the cell walls of the fibers contained a high amount of moisture, causing the cellulose microfibrils to be farther apart, and their alignment was not dense enough to provide the required resistance to axial tensile stress. Alongside this, the average diameter of the fiber was found to be 2.0 mm, and it is possible that due to the presence of cavities [54], the effective area to resist axial tension was not adequate enough. As a result, the tensile strength was recorded to be lower.

3.2 Workability of Fresh Concrete (Slump Test)

Figure 4 shows the slump value of concrete mixes investigated in this study. It can be observed that with the increase in OPBF content, the workability of fresh concrete has been reduced. The control case containing 0% OPBF showed the highest slump value (54.50 mm), and a reduction of 62.5% was observed for the mix containing the highest OPBF percentage (5%). In this study, the reduction in slump could be attributed to water absorption (29.54%) of OPBF, despite been considerably lower compared to 44.7% reported by Momoh *et al.* [52]. Similar reductions in workability with the inclusion of natural fibers have been reported in prior studies [64, 66], often attributed to the increased surface area and the disruptive effect of fibers on the cement matrix. The observed decline in slump is likely influenced by both the increased surface area and the mechanical obstruction caused by the fibers during flow, consistent with findings from other fiber-reinforced systems [67, 68]. A comparable trend was also reported by De Silva and Naveen (2024), who observed reduced workability in mortar containing RHA and coconut coir [59].

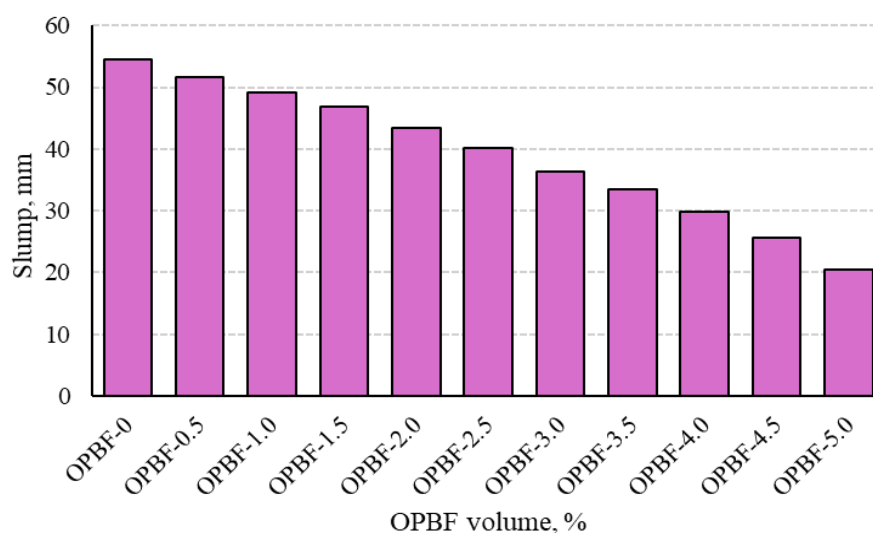


Figure 4 Slump of mixes containing varying amounts of OPBF (0%–5%).

3.3 Compressive Strength of the Hardened Concrete Cubes

Figure 5 presents the compressive strength of the concrete mixes studied at 28 days of curing, and the mean values with expanded uncertainty (U) at the 95% confidence level ($k = 2$) are also presented in Table 5. It is observed that compressive strength increased with the increase in OPBF content until it reached 2.5%, after which it decreased as the OPBF content increased further. This trend is consistent with many natural fiber studies. For example, More and Subramanian (2022) found that coir at 2% volume fraction improved compressive strength by about 15.9%, while steel, jute, and sisal gave similar improvements at around 1.5% dosage, with strength decreasing at higher levels due to poor fiber dispersion and void formation [69]. Varghese and Unnikrishnan (2023) likewise reported that coconut fiber concretes showed strength gains up to an optimum dosage before falling off, and Jin *et al.* (2023) observed a similar pattern in foamed concretes reinforced with coir–basalt hybrids, where optimum strength occurred at 0.3% fiber content [70, 71]. However, Momoh *et al.* observed a decreasing trend in compressive strength as the OPBF content increased in plain cement concrete and explained that the low radial strength of OPBF fibers, dynamic instability due to water absorption, and the alkaline degradation of fibers are among the contributing factors in reducing compressive strength [52]. For the current study, RHA has been incorporated as a SCM, and it might have positively impacted compressive strength up to a certain level of OPBF incorporation (2.5%). This can be attributed to RHA refining the interfacial transition zone (ITZ) around the fibers, making the fiber-matrix bond denser. However, when the OPBF content exceeded the optimum level, the amount of RHA used was not sufficient to fill the micro voids around the fibers. As a result, the fiber-matrix bond may have become concentrated in certain areas of the matrix, leading to heterogeneity in strength, which created weak zones and ultimately decreased overall cube strength. In addition, the reduced workability observed at higher fiber contents may have limited proper compaction, contributing to incomplete densification and the presence of internal voids, which further compromised strength. Although the ITZ was not directly observed in this study, this interpretation is based on similar findings reported in the literature.

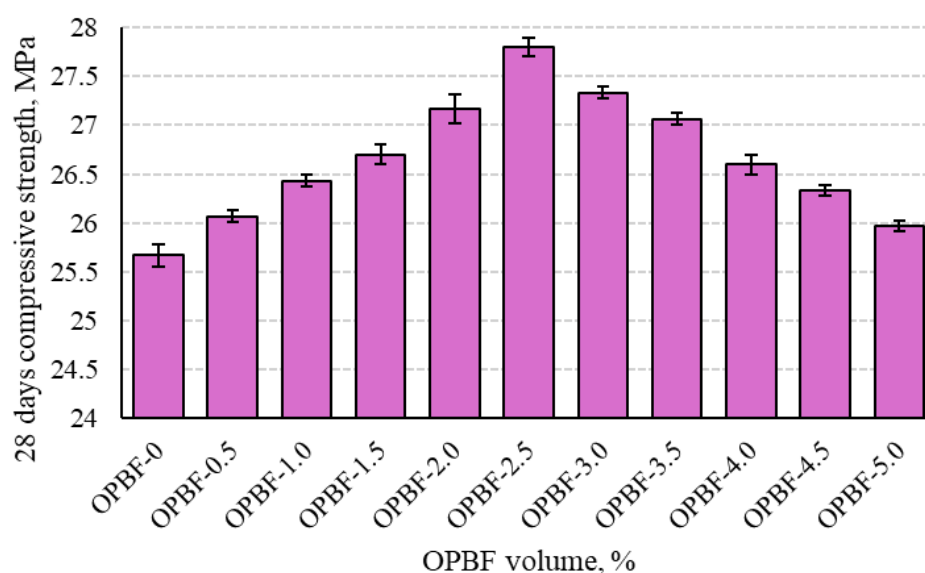


Figure 5 28-Days compressive strength of RHA concrete composed of different OPBF ratios (0%–5%).

Table 5 Compressive Strength of RHA Concrete with Varying OPBF Content (Mean $\pm$ U, 95% CL, k = 2).

OPBF volume (%)	0	0.5	1	1.5	2	2.5	3	3.5	4	4.5	5
28-days Compressive strength (MPa)	25.67 $\pm$ 0.13	26.07 $\pm$ 0.07	26.43 $\pm$ 0.07	26.70 $\pm$ 0.12	27.17 $\pm$ 0.18	27.80 $\pm$ 0.12	27.33 $\pm$ 0.07	27.07 $\pm$ 0.07	26.60 $\pm$ 0.12	26.33 $\pm$ 0.07	25.97 $\pm$ 0.07

To further verify the effect of OPBF incorporation on the compressive strength of concrete, a statistical evaluation was conducted, where a single factor analysis of variance (ANOVA) and Tukey's honest significant difference (HSD) analysis were performed using the 28-day compressive strength results obtained from the three concrete cubes made for each mix studied.

The ANOVA yielded a highly significant F-statistic of 162.04, which exceeds the critical F-value (2.297), with an extremely low p-value (3.236×10^{-18}). Given that the p-value is well below the significance threshold of 0.05, we conclude that there are statistically significant differences among the average 28 days compressive strength of the mixes studied. These findings suggest that incorporating OPBF had a strong influence on the compressive strength. As the ANOVA analysis revealed a strong influence of OPBF incorporation on concrete compressive strength, a Tukey HSD analysis was conducted to determine how significantly the mean 28-day compressive strength differed between the mixes studied. The q-statistic in the Tukey HSD test is used to compare the differences between group means while accounting for multiple comparisons. A higher q-statistic indicates a larger difference between two groups, whereas a lower q-statistic suggests that the groups are more similar. While ANOVA revealed statistically significant differences in compressive strength between mixes, the statistical significance should be viewed as confirming trends rather than indicating practical performance differences.

Figure 6(a) illustrates the pairwise comparisons of different OPBF levels with respect to compressive strength. The green-colored comparisons suggest major performance differences, whereas the red regions indicate little to no difference. This visualization aids in identifying the most influential OPBF levels in the study. For example, when comparing the influence of incorporating 2.5% OPBF with the control case, the q-statistic value reached the highest value of 41.8, and the cell color was deep green, whereas comparing it to 5% inclusion resulted in the lowest value of 5.3, turning the cell color deep red. Moreover, for most of the comparison pairs involving 2.5% OPBF, the q-statistic was the highest, and the cell colors were green. As a result, it is viable to conclude that 2.5% OPBF is the most positively influential OPBF amount. From Figure 6(b), it can be understood that the differences found in the previously mentioned cases were significant, as the p-values were well below 0.05.

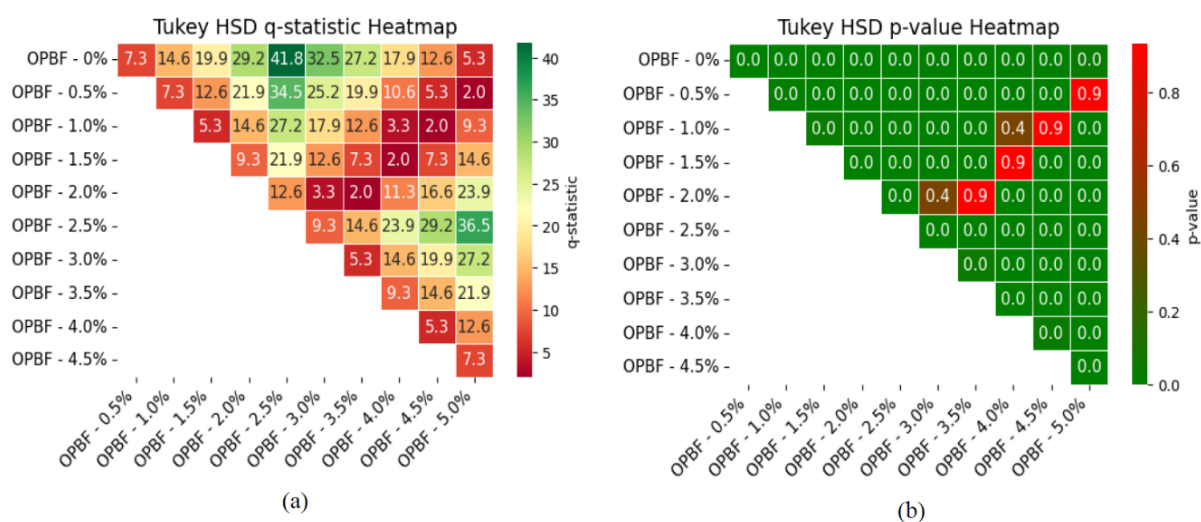


Figure 6 Tukey HSD analysis for compressive strength (a) q-statistic heatmap; (b) p-value heatmap.

To understand the effect of incorporating OPBF on the temporal development of compressive strength, a strength-normalized diagram was drawn (Figure 7). It is evident from the figure that incorporating different amounts of OPBF does not affect the compressive strength gain of RHA concrete over time. It can be observed from the figure that incorporating higher or lower OPBF amounts neither delays nor accelerates the hydration process.

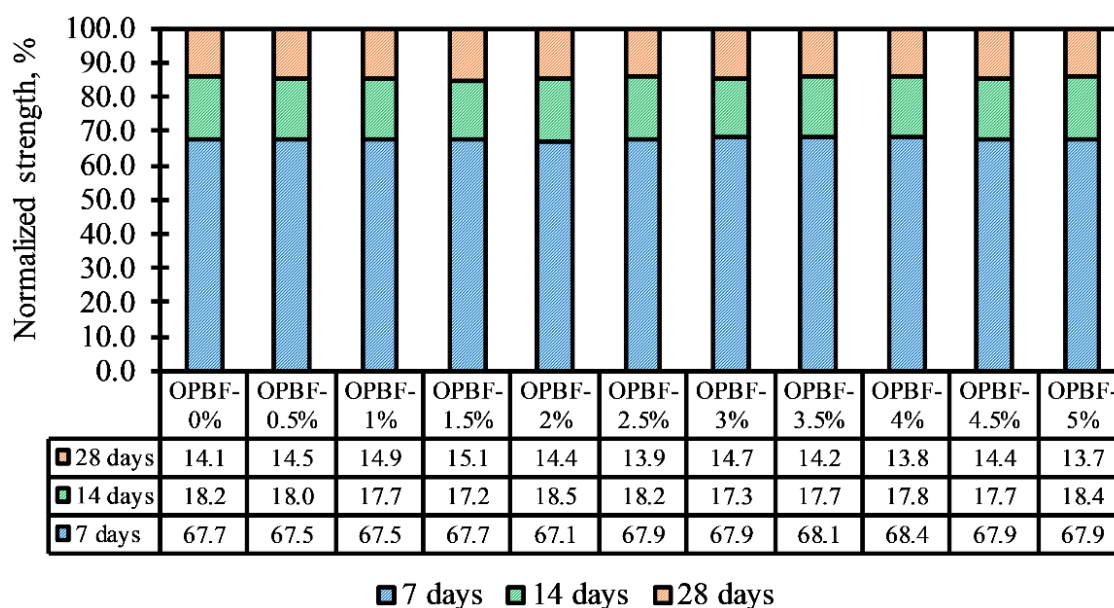


Figure 7 Compressive strength development of concrete incorporating different amounts of OPBF (0%–5%).

3.4 Flexural Strength of the Hardened Concrete Prisms

The summary of the average flexural strength of the concretes after two curing ages is shown in Figure 8, and the mean values with expanded uncertainty (U) at the 95% confidence level ($k = 2$) are also shown in Table 6. It can be observed that the flexural strength of RHA-blended concrete increased as the curing age increased. Similar to compressive strength, flexural strength also increased with the increase in OPBF content until it reached a certain level, which was 3%. The maximum flexural strength (5.8 MPa) was observed for the mixture containing 3.0% OPBF. A comparable study on RHA concrete with coconut coir fiber reported a lower optimum content of 0.5%, with the improvement attributed to the tensile properties of the fibers and the refinement of the fiber–matrix interfacial transition zone facilitated by RHA [59]. Beyond the optimum level, however, both the present study and previous work noted that increasing fiber content results in fiber agglomeration and weak bonding with the matrix, thereby reducing flexural strength. More and Subramanian (2022) found that steel, carbon, glass, coir, jute, and sisal fibers all provided flexural strength gains in the range of 12–16% at optimum dosages, emphasizing the crack-bridging effect of fibers [69]. Varghese and Unnikrishnan (2023) reported smaller improvements in coconut fiber concretes (about 2% increase at 0.5% dosage with longer fibers), highlighting the sensitivity of flexural performance to fiber type, size, and dosage [70]. Similarly, Jin *et al.* (2023) showed that hybrid natural fibers (coir–basalt) in foamed concretes achieved larger flexural strength increases (10–55% at 28 days), again underscoring the importance of fiber bridging in enhancing ductility [71]. Taken together, these comparisons confirm that the flexural strength behavior observed for OPBF

in RHA-blended concrete is consistent with the general trends reported for other natural fiber-reinforced concretes, while also identifying OPBF's optimum dosage at a higher level (3%).

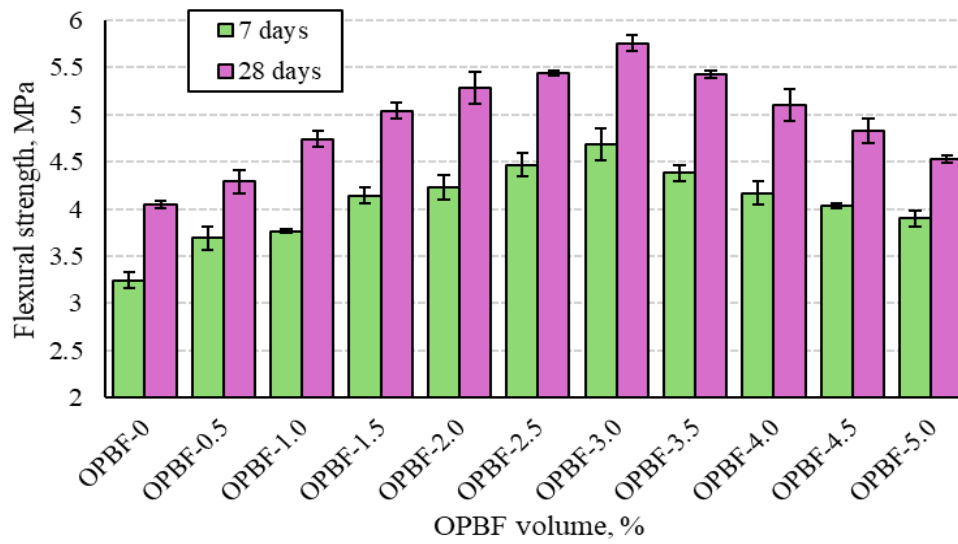


Figure 8 Flexural strength of RHA concrete composed of different OPBF ratios (0%–5%).

Table 6 Flexural Strength of RHA Concrete with Varying OPBF Content (Mean $\pm$ U, 95% CL, k = 2).

OPBF volume (%)	0	0.5	1	1.5	2	2.5	3	3.5	4	4.5	5
7-Days Flexural strength	3.24 $\pm$ 0.12	3.69 $\pm$ 0.18	3.77 $\pm$ 0.03	4.14 $\pm$ 0.12	4.23 $\pm$ 0.18	4.47 $\pm$ 0.18	4.68 $\pm$ 0.24	4.38 $\pm$ 0.12	4.17 $\pm$ 0.18	4.04 $\pm$ 0.03	3.90 $\pm$ 0.12
28-Days Flexural strength	4.05 $\pm$ 0.06	4.29 $\pm$ 0.18	4.74 $\pm$ 0.12	5.04 $\pm$ 0.12	5.28 $\pm$ 0.24	5.44 $\pm$ 0.04	5.76 $\pm$ 0.12	5.43 $\pm$ 0.06	5.10 $\pm$ 0.24	4.83 $\pm$ 0.18	4.53 $\pm$ 0.06

A statistical analysis, similar to the one conducted for compressive strength, was performed to verify the effect of OPBF on flexural strength. The ANOVA yielded a highly significant F-statistic of 52.08, which exceeds the critical F-value (2.85), with an extremely low p-value (8.37×10^{-8}). Given that the p-value is well below the significance threshold of 0.05, we conclude that there are statistically significant differences among the average 28 days flexural strength of the mixes studied. Figure 9 illustrates the pairwise comparisons and the statistical significance of different OPBF levels concerning flexural strength. The results indicate that incorporating 2.5%–3% OPBF significantly enhances flexural strength.

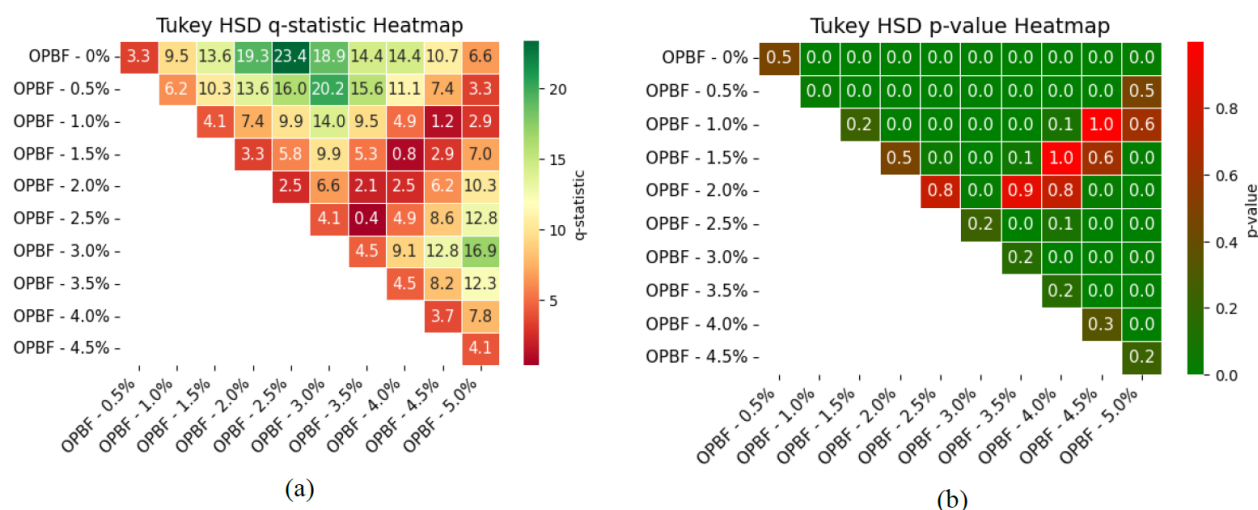


Figure 9 Tukey HSD analysis for flexural strength (a) q-statistic heatmap; (b) p-value heatmap.

4. Conclusion

This study investigated the effect of OPBF at varying dosages on the fresh and mechanical properties of RHA-blended concrete. The experimental program was conducted at the laboratory scale, with slump, compressive strength, and flexural strength evaluated at 7, 14, and 28 days. The scope of the work was limited to short-term mechanical performance and statistical analysis of results. Within the limits of this experimental program, the following conclusions can be drawn:

1. The OPBF used in this study had a moisture content of 12.44% and a water absorption capacity of 29.54%. This moderate absorption reflects limited pore availability for additional water uptake, which in turn influenced the workability of fresh concrete.
2. The measured tensile strength of OPBF (105.38 MPa) suggests that the fiber has sufficient mechanical capacity to contribute to stress redistribution and crack-bridging within the concrete matrix.
3. Compressive strength improved with OPBF addition up to 2.5%, beyond which strength declined. Statistical analysis (ANOVA and Tukey's HSD) identified 2.5% as the optimal OPBF dosage. Aligning with previous work, microstructural improvements such as ITZ densification are hypothesized to contribute to the strength gain. Of note, further research focusing on imaging and interface analysis will be analyzed to validate this mechanism.
4. Flexural strength increased with OPBF content, reaching a peak of 5.8 MPa at 3.0% fiber dosage and the observed reduction in flexural strength at higher fiber dosages may be

attributed to fiber agglomeration and reduced bonding efficiency. Future work will incorporate interface imaging and mechanical characterization to confirm this mechanism.

5. With compressive strength values reaching 27.8 MPa and flexural strength peaking at 5.8 MPa, the RHA – OPBF concrete developed in this study exceeds the minimum requirements of ASTM C90 (13.8 MPa) for load-bearing concrete masonry units and ASTM C129 (3.5 MPa) for non-loadbearing units, confirming its suitability for partition walls, precast panels, and other low-load-bearing elements in low-rise construction. However, the strength remains below the $\geq 30 - 40$ MPa thresholds of ACI 318 for structural members, positioning this material as a sustainable alternative for non-structural and low-load applications where enhanced crack resistance and eco-efficiency are desired.

Abbreviations

OPBF	Oil Palm Broom Fiber
RHA	Rice husk ash
OPC	Ordinary Portland Cement
SCM	Supplementary cementitious materials
ANOVA	Analysis of variance
Tukey's HSD	Tukey's Honest Significant Difference
POFA	Palm oil fuel ash
LOI	loss on ignition
XRF	X-ray fluorescence
FA	Fine aggregate
CA	Coarse aggregate
Cu	Coefficient of uniformity
Cc	Coefficient of curvature
DOE	Department of Environment
W/B	water-to-binder
ITZ	Interfacial transition zone
CO ₂	Carbon dioxide
wt%	Weight percent
U	Expanded uncertainty
CL	Confidence level
k	Coverage factor

Author Contributions

Taofiq O. Mohammed: Conceptualization, methodology, investigation, data curation, writing—original draft preparation. Ebenezer O. Fanijo: Methodology, writing—review and editing, visualization, supervision. Mohammad Zunaied Bin Harun: Software, validation. Jian Liu: Software, validation. All authors have read and agreed to the published version of the manuscript.

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Competing Interests

On behalf of all authors, the corresponding author states that there is no conflict of interest.

Data Availability Statement

Data will be made available upon request.

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